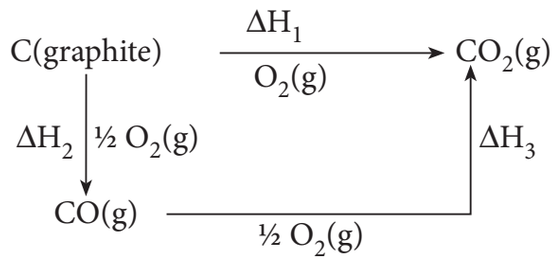
However, enthalpy for the oxidation of graphite to CO_2 and CO to CO_2 can easily be measured. For these conversions, the heat of combustion values are –393.5 kJ and – 283.5 kJ respectively.

From these data the enthalpy of combustion of graphite to CO can be calculated by applying Hess's law.

The reactions involved in this process can be expressed as follows



According to Hess law,

$$\Delta H_1 = \Delta H_2 + \Delta H_3$$

$$-393.5 \text{ kJ} = X - 283.5 \text{ kJ}$$

$$X = -110.5 \text{ kJ}$$

7.9 Lattice energy ($\Delta H_{lattice}$)

Lattice energy is defined as the amount of energy required to completely remove the constituent ions from its crystal lattice to an infinite distance from one mole of crystal. It is also referred as lattice enthalpy.

$$NaCl(s) \rightarrow Na^+(g) + Cl^-(g)$$

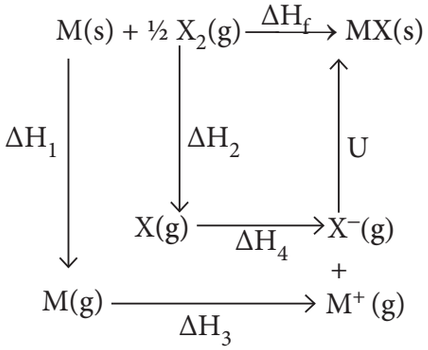
$$\Delta H_{lattice} = + 788 \text{ kJ mol}^{-1}$$

From the above equation it is clear that 788 kJ of energy is required to separate Na^+ and Cl^- ions from 1 mole of NaCl.

Born - Haber cycle

The Born–Haber cycle is an approach to analyse reaction energies. It was named after two German scientists Max Born and Fritz Haber who developed this cycle. The cycle is concerned with the formation of an ionic compound from the reaction of a metal with a halogen or other non-metallic element such as oxygen.

Born–Haber cycle is primarily used in calculating lattice energy, which cannot be measured directly. The Born–Haber cycle applies Hess's law to calculate the lattice enthalpy. For example consider the formation of a simple ionic solid such as an alkali metal halide MX, the following steps are considered.



ΔH_1 - enthalpy change for the sublimation M(s) to M(g)

ΔH_2- enthalpy change for the dissociation of $\frac{1}{2} X_2$ (g) to X(g)

ΔH_3- Ionisation energy for M(g) to M^+(g)

ΔH_4 - electron affinity for the conversion of X(g) to X^-(g)

U - the lattice enthalpy for the formation of solid MX

ΔH_f - enthalpy change for the formation of solid MX directly form elements

According to Hess's law of heat summation

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